

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes',
'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri',
'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets',
'seaice', 'taxi', 'tips', 'titanic']

dataset=sns.load_dataset('titanic')
dataset.head()

{"summary":{"\n  \"name\": \"dataset\", \n  \"rows\": 891, \n
  \"fields\": [\n    {\n      \"column\": \"survived\", \n
  \"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
0, \n        \"min\": 0, \n        \"max\": 1, \n
  \"num_unique_values\": 2, \n        \"samples\": [\n          1, \n
0 \n        ], \n        \"semantic_type\": \"\", \n
  \"description\": \"\" \n    }, \n    {\n      \"column\":
  \"pclass\", \n        \"properties\": {\n          \"dtype\": \"number\", \n
  \"std\": 0, \n          \"min\": 1, \n          \"max\": 3, \n
  \"num_unique_values\": 3, \n          \"samples\": [\n            3, \n
1 \n          ], \n          \"semantic_type\": \"\", \n
  \"description\": \"\" \n    }, \n    {\n      \"column\":
  \"sex\", \n        \"properties\": {\n          \"dtype\": \"category\", \n
  \"num_unique_values\": 2, \n          \"samples\": [\n
  \"female\", \n          \"male\" \n        ], \n
  \"semantic_type\": \"\", \n        \"description\": \"\" \n    }
  ], \n    {\n      \"column\": \"age\", \n        \"properties\": {\n
  \"dtype\": \"number\", \n        \"std\": 14.526497332334044, \n
  \"min\": 0.42, \n        \"max\": 80.0, \n        \"num_unique_values\":
88, \n        \"samples\": [\n          0.75, \n          22.0 \n
        ], \n        \"semantic_type\": \"\", \n
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  \"std\": 1, \n          \"min\": 0, \n          \"max\": 8, \n
  \"num_unique_values\": 7, \n          \"samples\": [\n            1, \n
0 \n          ], \n          \"semantic_type\": \"\", \n
  \"description\": \"\" \n    }, \n    {\n      \"column\":
  \"parch\", \n        \"properties\": {\n          \"dtype\": \"number\", \n
  \"std\": 0, \n          \"min\": 0, \n          \"max\": 6, \n
  \"num_unique_values\": 7, \n          \"samples\": [\n            0, \n
1 \n          ], \n          \"semantic_type\": \"\", \n
  \"description\": \"\" \n    }, \n    {\n      \"column\":
  \"fare\", \n        \"properties\": {\n          \"dtype\": \"number\", \n
  \"std\": 49.693428597180905, \n          \"min\": 0.0, \n          \"max\":
512.3292, \n          \"num_unique_values\": 248, \n          \"samples\":
[\n            11.2417, \n            51.8625 \n          ], \n

```

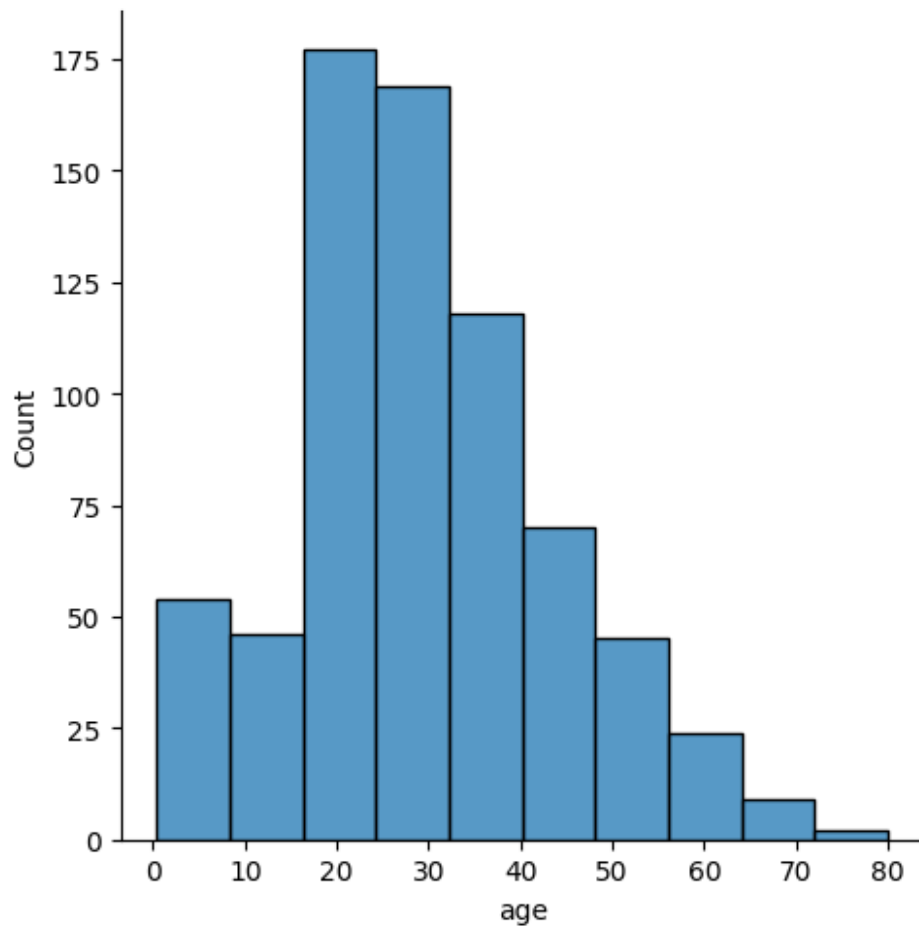
```

{"semantic_type": "",
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 "column": "embarked",
 "properties": {
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 },
 "semantic_type": "",
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   "samples": ["Third", "First"]
 },
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   "samples": ["man", "woman"]
 },
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 "properties": {
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   "samples": [false, true]
 },
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   "num_unique_values": 7,
   "samples": ["C", "E"]
 },
 "semantic_type": "",
 "description": "",
 "column": "embark_town",
 "properties": {
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   "num_unique_values": 3,
   "samples": ["Southampton", "Cherbourg"]
 },
 "semantic_type": "",
 "description": "",
 "column": "alive",
 "properties": {
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   "num_unique_values": 2,
   "samples": ["yes", "no"]
 },
 "semantic_type": "",
 "description": "",
 "column": "alone",
 "properties": {
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   "num_unique_values": 2,
   "samples": [true, false]
 },
 "semantic_type": "",
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}]
}, {"type": "dataframe", "variable_name": "dataset"}

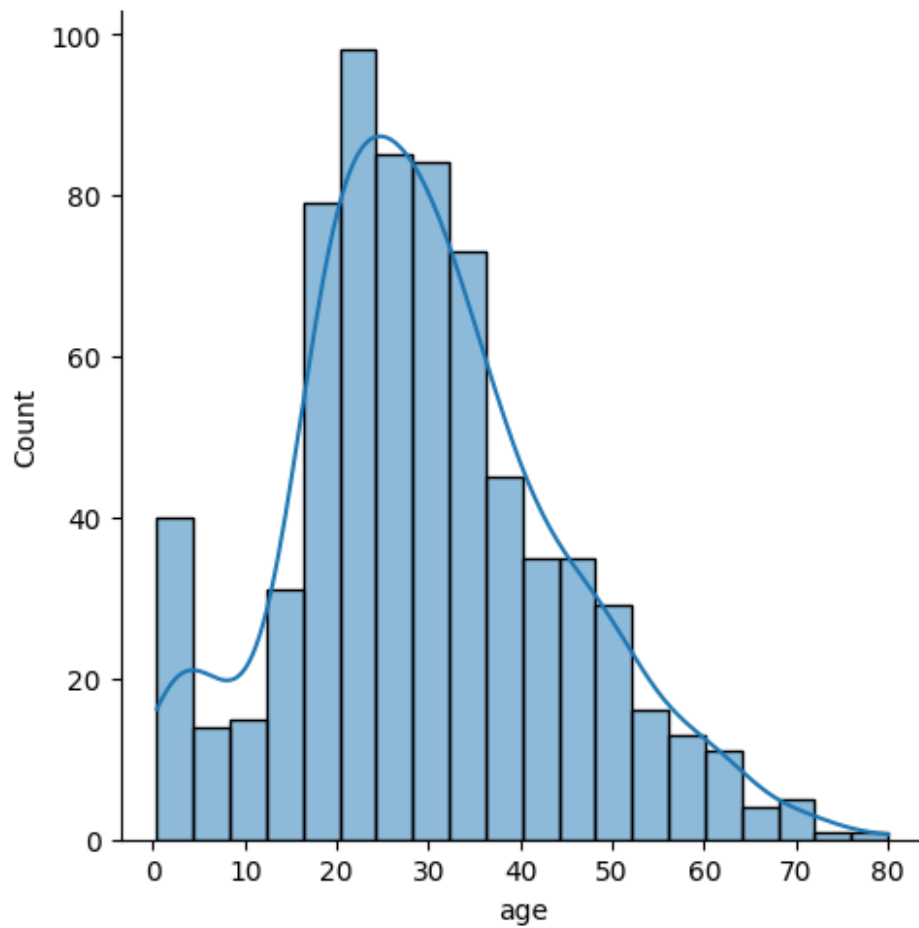
```

```
sns.displot(x=dataset['age'])
```

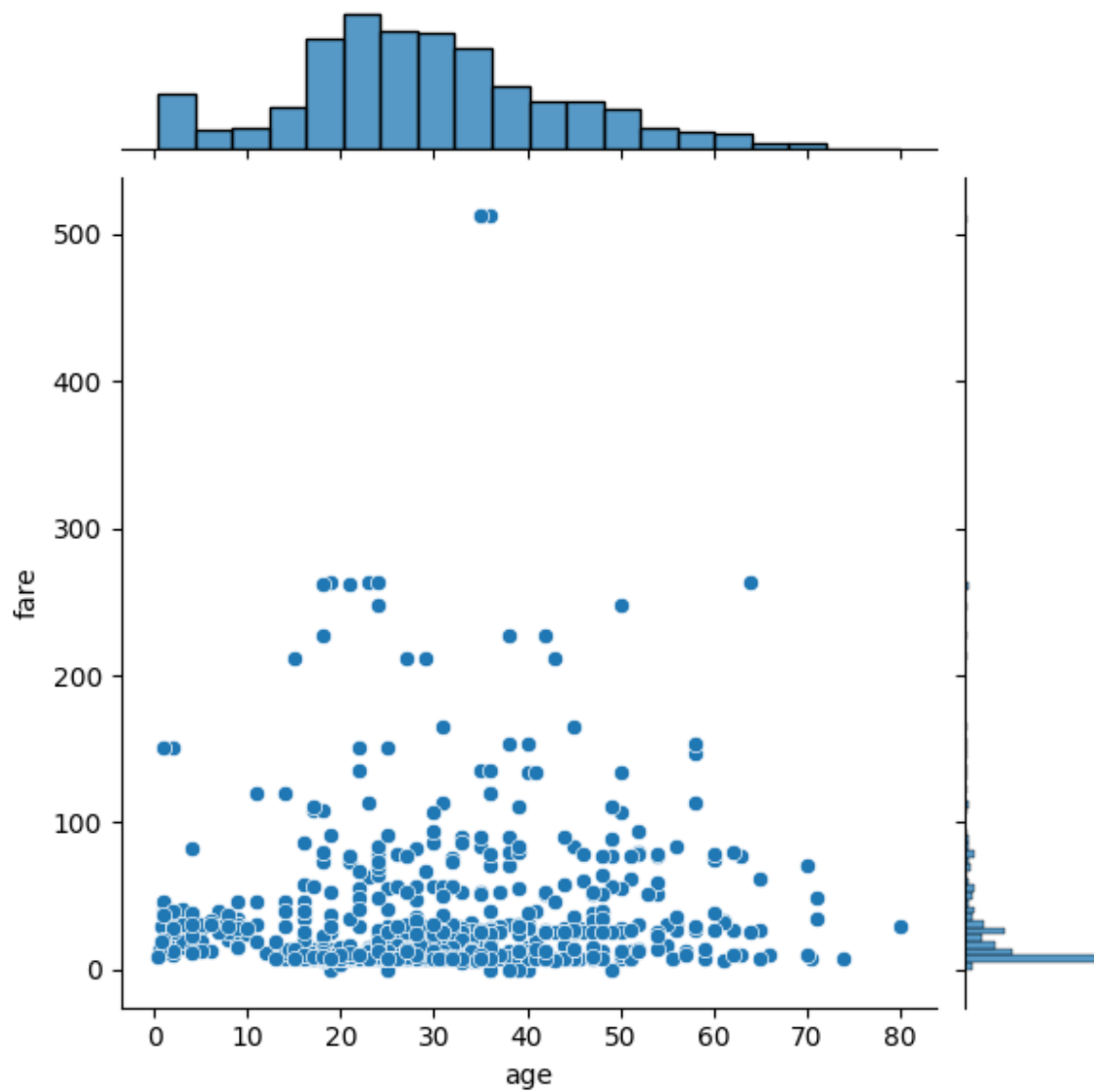
```
<seaborn.axisgrid.FacetGrid at 0x7a23781b1850>
```



```
sns.displot(x=dataset['age'],kde=True)  
<seaborn.axisgrid.FacetGrid at 0x7a2378366750>
```

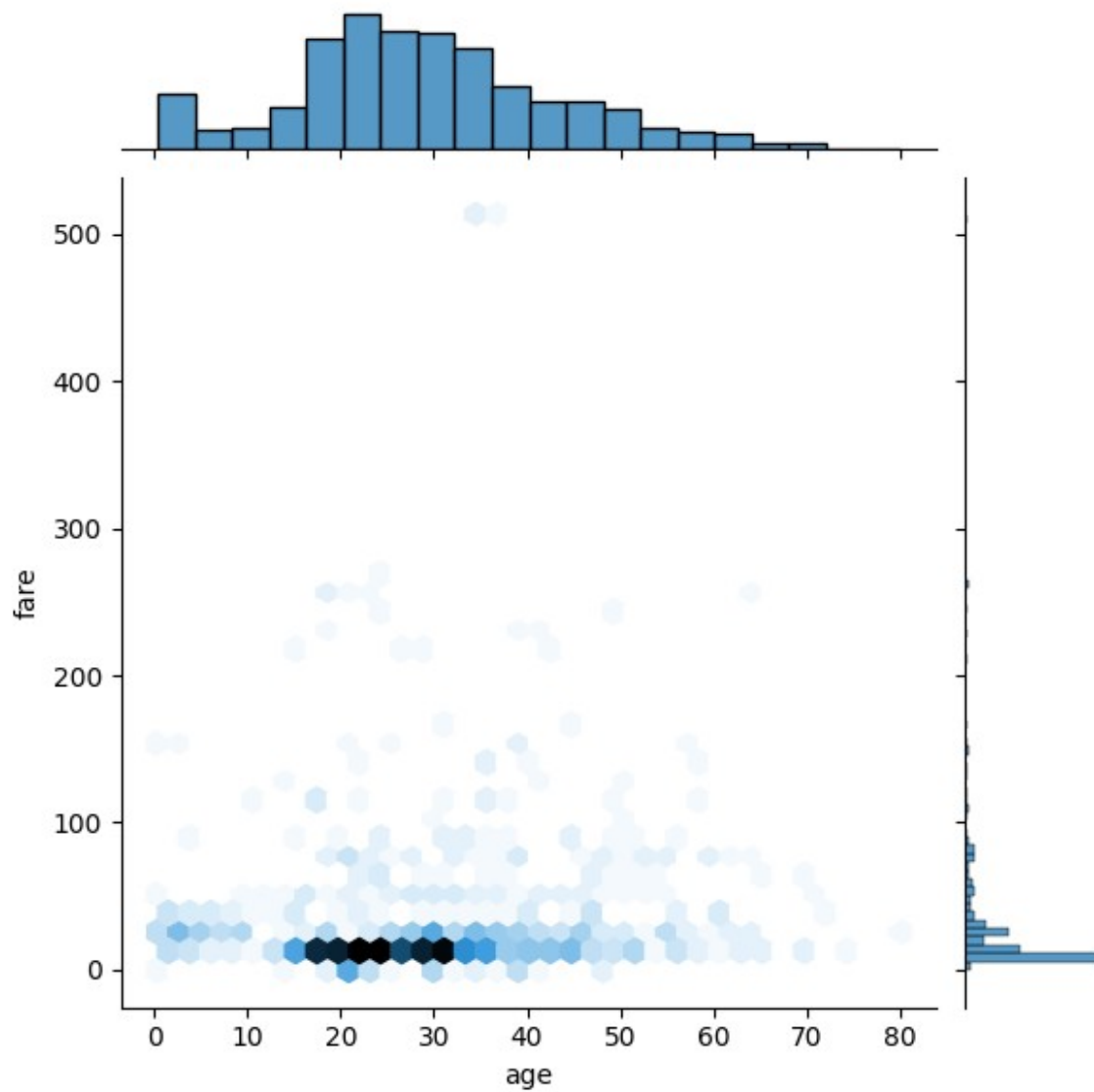


```
sns.jointplot(x=dataset['age'],y=dataset['fare'],kind='scatter')  
<seaborn.axisgrid.JointGrid at 0x7a2375ec7410>
```



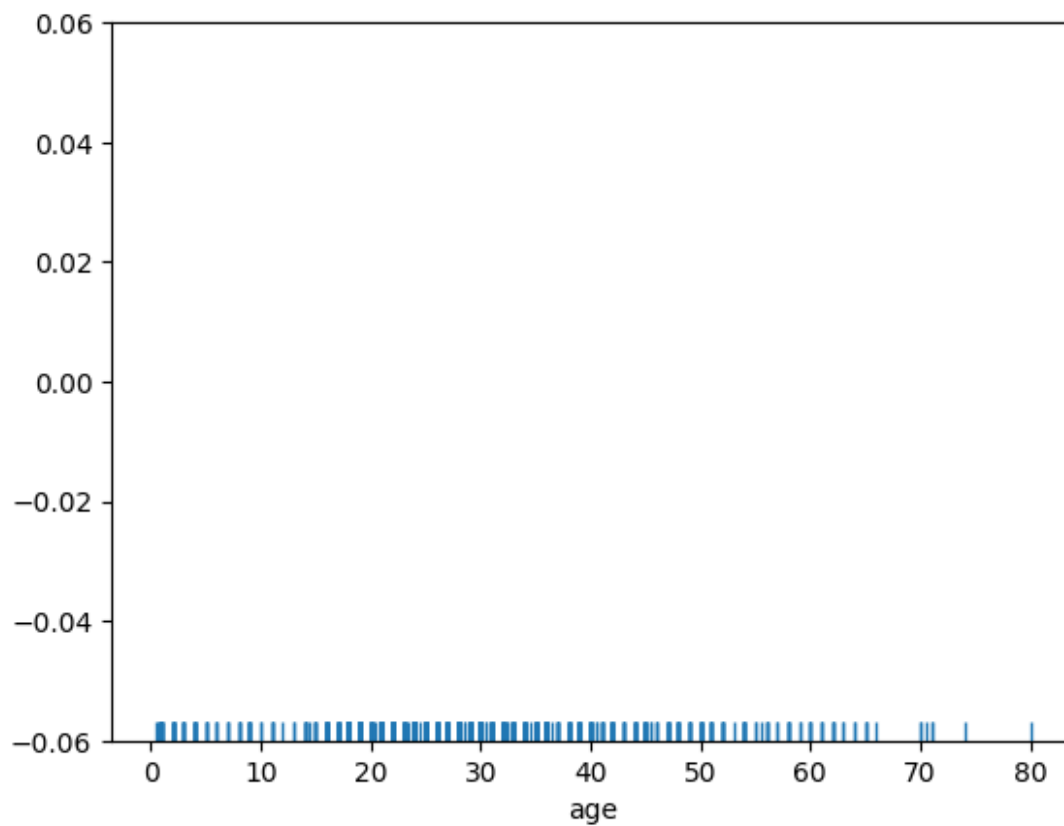
```
sns.jointplot(x=dataset['age'],y=dataset['fare'],kind='hex')
```

```
<seaborn.axisgrid.JointGrid at 0x7a2373abfa50>
```

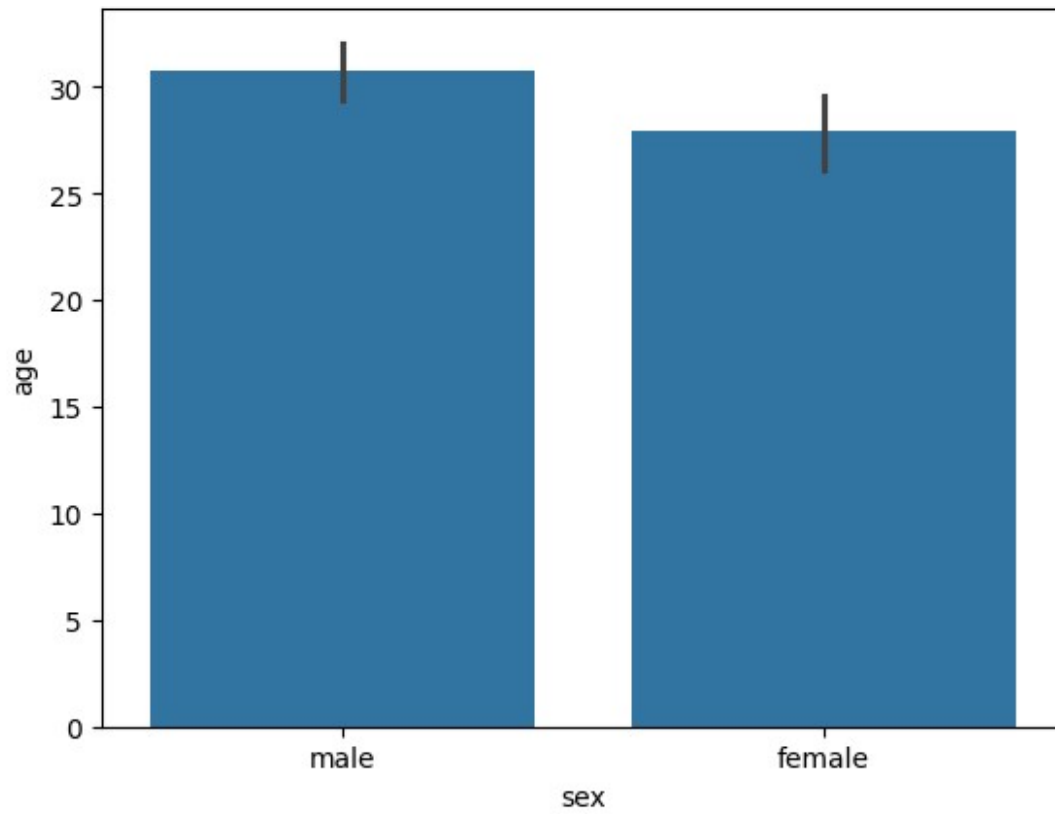


```
sns.rugplot(x=dataset['age'])
```

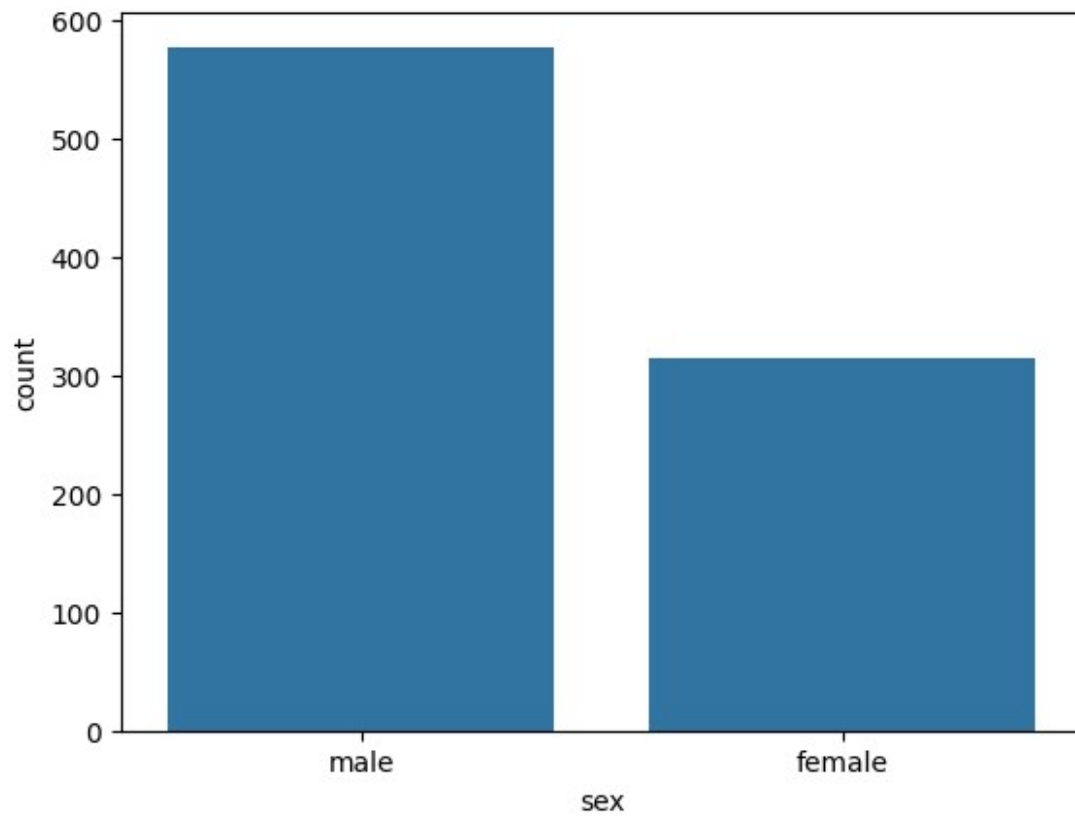
```
<Axes: xlabel='age'>
```



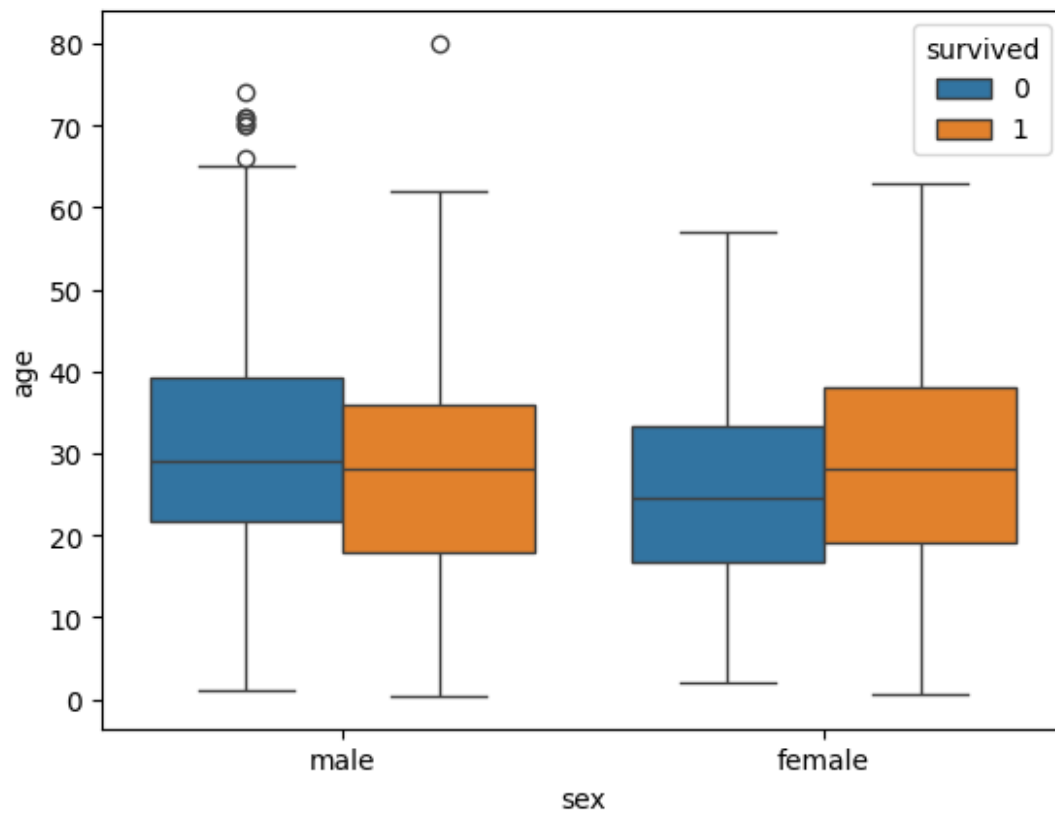
```
sns.barplot(x=dataset['sex'],y=dataset['age'])  
<Axes: xlabel='sex', ylabel='age'>
```



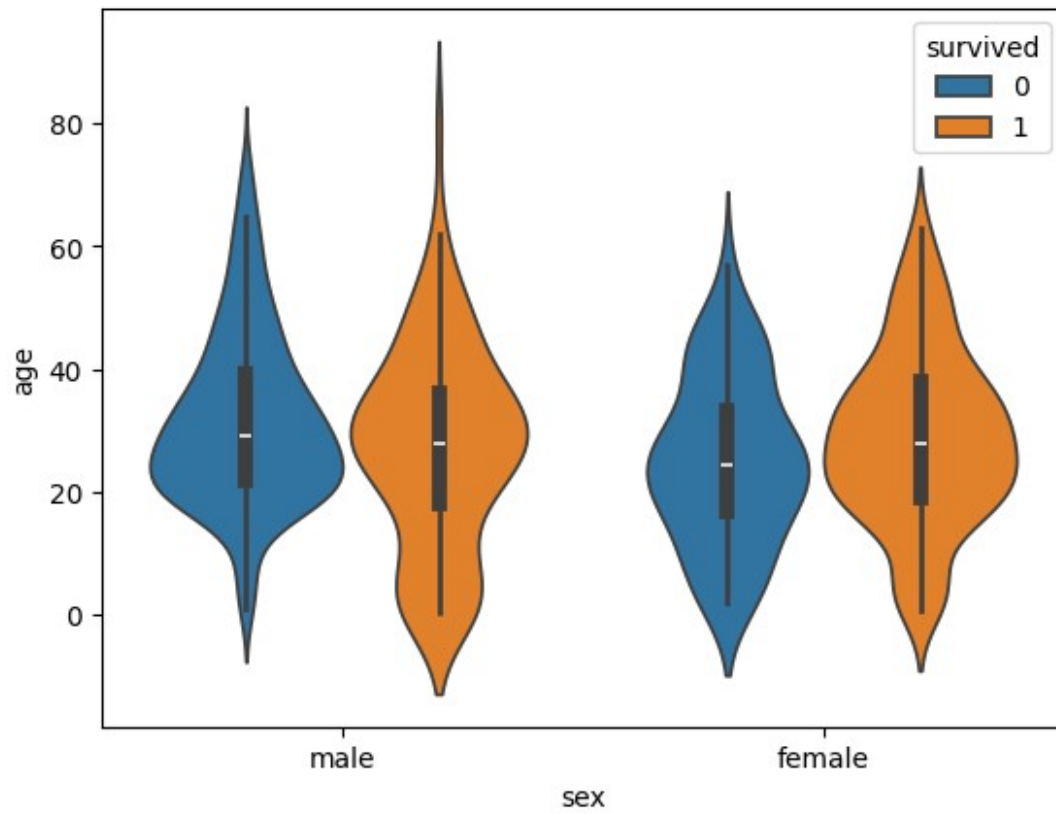
```
sns.countplot(x=dataset['sex'])  
<Axes: xlabel='sex', ylabel='count'>
```

```
sns.boxplot(x='sex', y='age', data=dataset, hue='survived')  
<Axes: xlabel='sex', ylabel='age'>
```



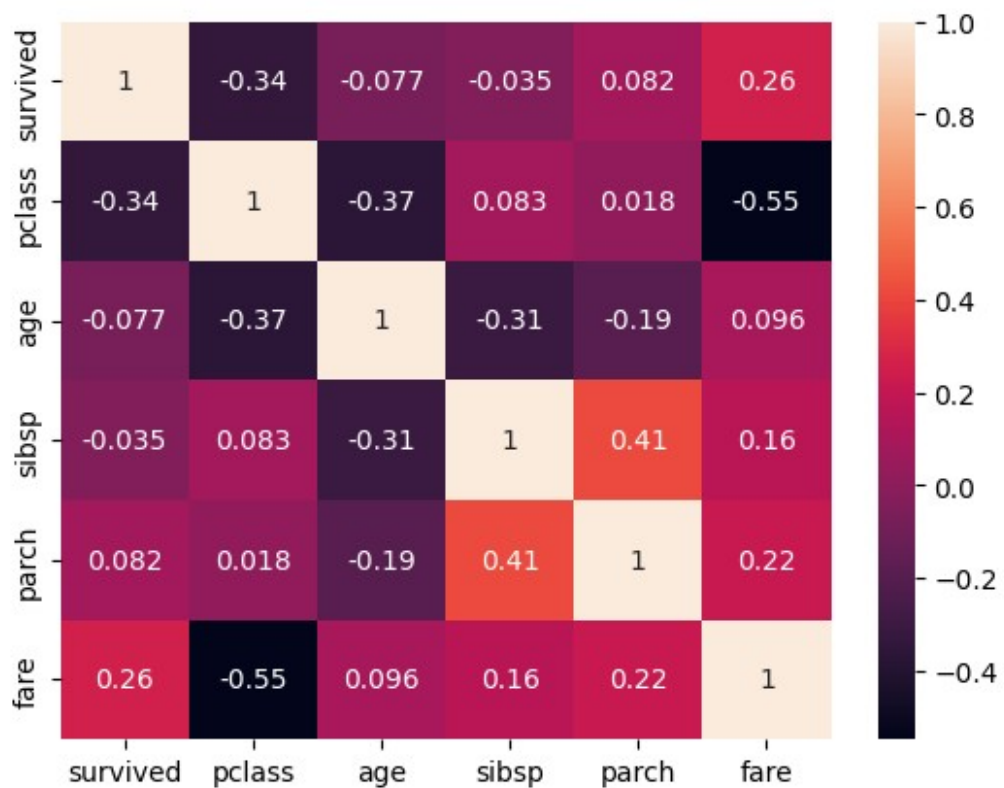
```
sns.violinplot(x='sex', y='age', data=dataset, hue='survived')  
<Axes: xlabel='sex', ylabel='age'>
```



#heatmaps

```
corr = dataset.select_dtypes(include=['number']).corr()  
sns.heatmap(corr, annot=True)
```

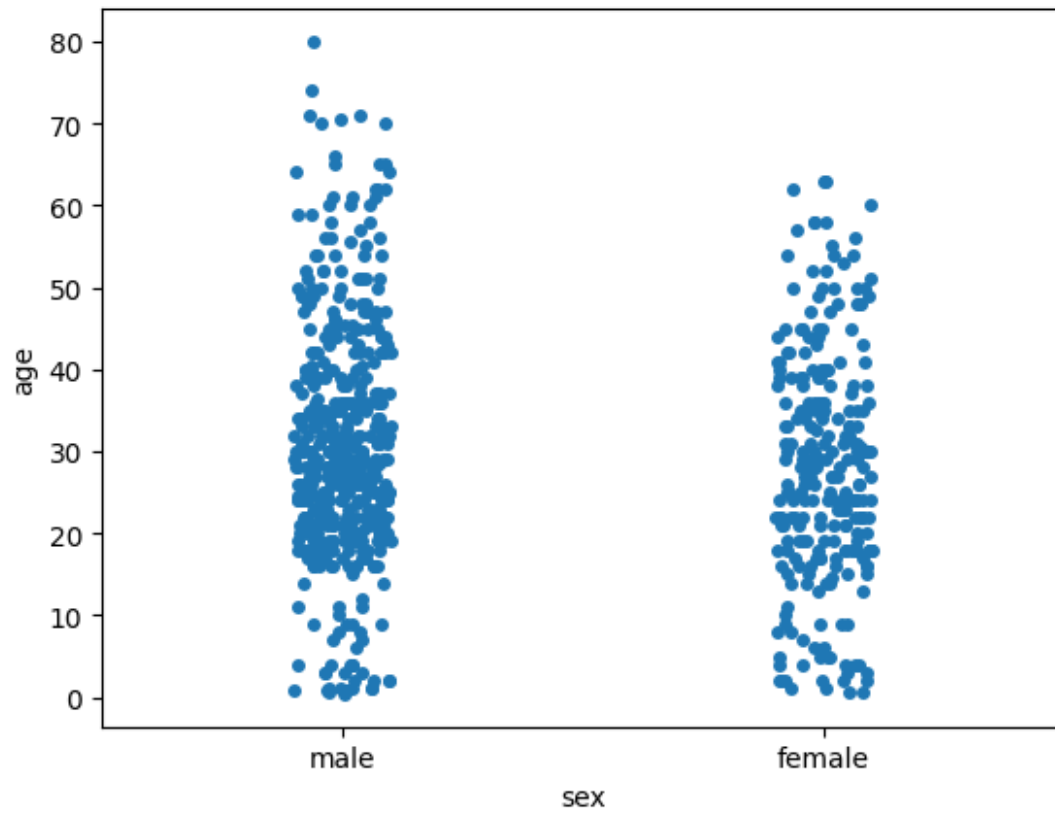
<Axes: >



```
#advanced plotting
```

```
sns.stripplot(x='sex', y='age', data=dataset, jitter=True)
```

```
<Axes: xlabel='sex', ylabel='age'>
```



```
sns.swarmplot(x='sex', y='age', data=dataset)  
<Axes: xlabel='sex', ylabel='age'>
```

